

BACKEND AND ADMIN PANEL

Implementation Requirements

A launch-ready handoff covering checkout, wallet, subscriptions, orders, serviceability, invoices and customer notifications.

FRONTEND STATUS: COMPLETE

BACKEND STATUS: ITEMS IN THIS DOCUMENT ARE PENDING

Prepared for the Atulyash backend team
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Checkout, delivery charges and wallet recharge

The backend must be the final authority for every monetary value. The frontend can explain a charge, but the cart, checkout and placed order must all return the same server-calculated amount.

01 Authoritative delivery-charge calculation

Calculate delivery charges on the server and return the rule used in the cart and order response.

- First Atulyash Experience delivery: free.
- Regular scheduled subscription delivery: free.
- Extra atta attached to the same scheduled subscription delivery: free.
- Separate one-time order: 2-6 kg = INR 40; 7-10 kg = INR 30; 11-40 kg = free.
- Above 40 kg: do not auto-confirm; return a customer-care approval state.

```
{
  "subtotal": 480,
  "discount": 100,
  "delivery_charge": 40,
  "wallet_debit": 420,
  "total": 420,
  "delivery_charge_reason": "ONE_TIME_2_TO_6_KG"
}
```

02 Four-delivery wallet requirement

A weekly plan can start only when the wallet covers one complete subscription month - four scheduled deliveries. Debit the wallet delivery by delivery, not as one upfront monthly debit.

```
{
  "available_balance": 1100,
  "price_per_delivery": 300,
  "minimum_deliveries_required": 4,
  "minimum_wallet_required": 1200,
  "shortfall": 100,
  "can_start_subscription": false
}
```

03 Razorpay wallet-recharge contract

The recharge initiation response must contain everything required to open Razorpay Checkout. Credit the wallet only after server-side signature verification.

- Return the Razorpay Key ID, Razorpay Order ID, amount in paise, currency and internal recharge ID.
- Verify razorpay_payment_id, razorpay_order_id and razorpay_signature on the server.
- Make verification idempotent so a retry cannot credit the wallet twice.

```
{
  "razorpay_key_id": "rzp_test_xxxxx",
  "razorpay_order_id": "order_xxxxx",
  "amount_in_paise": 100000,
  "currency": "INR",
  "wallet_recharge_id": 123
}
```

Subscription management

Customers must be able to change an active plan safely without exposing internal exceptions or creating conflicting delivery schedules.

04 Fix the subscription-change server exception

Changing a subscription or delivery schedule can currently return an unsupported operand / NoneType error. Validate nullable values before calculations and return structured field errors instead of implementation details.

No internal Python exception text should ever be returned to the customer interface.

05 Add a delivery weekday and effective-date API

Provide a supported mutation for changing the regular delivery weekday or the date from which a new schedule becomes effective.

- Validate delivery-route availability and milling cut-off rules.
- Do not modify deliveries that are already locked for production.
- Return the confirmed weekday, effective date and next delivery date.

```
PATCH /subscription/subscription_plan/{subscription_id}/schedule/  
  
{  
  "delivery_day": "Thursday",  
  "effective_from": "2026-08-20"  
}
```

06 Stabilise subscription pack updates

Verify the existing update-pack route and return the complete updated plan after a successful change.

- Expected endpoint: POST /subscription/subscription_plan/{id}/update-pack/
- Return subscription pack, weekly cycle, per-delivery price and minimum wallet requirement.
- Recalculate future unlocked deliveries without altering completed orders.

07 Enforce whole-kilogram weekly cycles

Monthly plans that do not divide equally across four weeks must use alternating whole-kilogram deliveries. The public API already returns the correct examples; the admin panel must preserve them.

- 10 kg/month: 3, 2, 3, 2 kg.
- 14 kg/month: 4, 3, 4, 3 kg.
- 18 kg/month: 5, 4, 5, 4 kg.
- Reject 0.5 kg weekly quantities.

Orders, modifications and invoices

The customer account already has the frontend entry points. The backend must provide complete order totals, modification eligibility and official invoice data.

08 One-time order modification API

Allow eligible one-time orders to change pack size, quantity, delivery date and delivery address before the fulfilment cut-off.

- Return which fields are still modifiable and the modification deadline.
- Reprice the order and adjust the wallet safely after every accepted change.
- Reject changes to orders already locked for milling, packing or dispatch.

```
{
  "can_modify": true,
  "modifiable_fields": [
    "product_pack", "quantity",
    "delivery_date", "customer_address"
  ],
  "modification_cutoff": "2026-08-15T18:00:00+05:30"
}
```

09 Official invoices and transaction history

Return invoice metadata with completed orders and expose a customer ledger covering every wallet movement.

- Order response: subtotal, discount, delivery charge, total, wallet debit and payment status.
- Invoice response: invoice number and secure PDF URL.
- Ledger: recharges, order debits, refunds, credits, payment references and balance after each transaction.
- Use pagination and stable ordering for the transaction-history API.

10 Prevent zero-value paid orders

Order list and detail responses must contain the actual order total after wallet payment. Do not return zero unless the final payable amount was genuinely zero.

The amount shown in Orders, the printable receipt, invoice and wallet ledger must reconcile exactly.

Serviceability, notifications and admin controls

Google can suggest an area in the frontend, but only Atulyash data can decide whether that PIN-code and area combination is serviceable.

11 PIN-code and area serviceability API

Make serviceability depend on the selected PIN-code and area combination, not on Google results alone.

- Return serviceable status and available delivery days.
- Return a clear non-serviceable reason for customer display.
- Use normalised PIN-code and area identifiers to avoid spelling mismatches.

```
GET /pincodes/serviceability/?pincode=110017&area;=Saket

{
  "pincode": "110017",
  "area": "Saket",
  "is_serviceable": true,
  "available_delivery_days": ["Tuesday", "Thursday", "Saturday"]
}
```

12 Admin serviceability controls

The admin panel must support adding, editing, activating and deactivating both PIN codes and areas.

- Manage areas under each PIN code.
- Configure delivery days and any region-specific delivery rules.
- Prevent deletion when active subscriptions depend on a serviceability record; allow safe deactivation instead.

13 Wallet and delivery notification sequence

Trigger customer notifications at the key points in a four-delivery wallet cycle.

- Immediately after recharge.
- After the second delivery and after the third delivery.
- Approximately one day before the fourth and final funded delivery.
- After the fourth delivery.
- When the next delivery is held due to insufficient balance.

14 Late-recharge behaviour

Recharge before the order cut-off should retain the upcoming delivery. Recharge after the cut-off should resume from the next available delivery.

15 Admin-configurable subscription policy

Allow authorised administrators to configure the minimum funded deliveries, notification timings and messages, order cut-off time, pause/resume rules and optional auto-recharge activation.

ACCEPTANCE CHECKLIST

Definition of backend-ready

The backend work is complete only when the following checks pass in test mode and can be repeated without creating duplicate carts, payments, wallet credits or orders.

AREA	PASS CONDITION
Cart totals	Cart, checkout, placed order, receipt and wallet ledger show identical monetary values.
Delivery rules	Every delivery-charge scenario returns the correct amount and a stable reason code.
Weekly wallet	A plan cannot start below four-delivery cover; each delivery is debited separately.
Razorpay	Recharge creation, verification and retries are idempotent; no duplicate credit is possible.
Subscriptions	Pack changes, skip/unskip, vacation and schedule changes return customer-safe responses.
Order changes	Eligibility and cut-off rules are enforced and accepted changes are repriced.
Invoices	Invoice URL, order totals and transaction ledger reconcile.
Serviceability	PIN-code plus area is checked against Atulyash records and admin activation state.
Notifications	The complete four-delivery sequence is testable and admin configurable.

Frontend handoff status: complete. The customer-facing screens, validation states and error presentation are ready to consume the backend contracts described in this document.